Analyzing Running Performance: The Impact of Training Frequency, Break Periods, and Running Environment

1. Motivation

Running performance evolves over time depending on training consistency, frequency, and external conditions. Many runners experience irregular training patterns, including extended breaks, which can significantly affect performance. Understanding how these interruptions influence performance decline and recovery is essential for designing effective training strategies.

In addition, running environment may play a role in performance outcomes. Treadmill (indoor) and outdoor running differ in terms of conditions such as terrain, pacing, and environmental resistance. These differences may influence performance metrics and training efficiency.

This project aims to analyze personal running data to understand how training frequency, inactivity periods, and running environment affect performance over time.

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2. Research Questions and Sub-Questions

Main Question

How do training frequency and inactivity periods affect running performance over time?

Sub-Questions

- How does training frequency relate to performance improvement?
- How much performance declines after long breaks from running?
- How long does it take to recover to previous performance levels after a break?
- Do treadmill and outdoor runs differ in terms of performance metrics such as pace and heart rate?
- Does running environment influence recovery speed and improvement trends?

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3. Hypotheses

H1: Higher training frequency is associated with improved running performance.

H2: Longer inactivity periods lead to greater performance decline.

H3: Recovery time increases with the length of the break period.

H4: Running environment (treadmill vs outdoor) significantly affects performance metrics.

H0 (Null): Training frequency, inactivity periods, and running environment have no significant effect on running performance.

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4. Data Description

Dataset: My personal running activity dataset collected via a fitness tracking device.

Variables:

- Distance
- Duration
- Pace
- Heart rate
- Date and time

Derived Variables:

- Weekly training frequency
- Time gaps between runs
- Identification of long break periods
- Running environment classification (treadmill vs outdoor)

Processed outputs will include structured time-series and aggregated datasets for analysis.

Garmin Running Dataset

Activity Type	Date	Favorite	Title	Distance	Calories	Time	Avg HR	Max HR	Aerobic TE	Avg Run Cadence	Max Run Cadence	Avg Pace	Best Pace	Total Ascent	Total Descent	Avg Stride Length	Avg Vertical Ratio	Avg Vertical Oscillation	Avg Ground Contact Time	Avg GAP	Normalized Power® (NP®)	Training Stress Score®	Avg Power	Max Power	Steps	Body Battery Drain	Min Temp	Decompression	Best Lap Time	Number of Laps	Max Temp	Moving Time	Elapsed Time	Min Elevation	Max Elevation
Running	2026-03-14 16:20:29	FALSE	Tuta Running	5.08	451	00:32:11	164	177	3.9	147	152	6:20	5:40	38	41	1.07	8.9	9.4	320	6:21	362	0.0	359	465	4,736	-11	--	No	00:02:27.5	8	--	00:32:10	00:32:11	128	140
Treadmill Running	2026-03-08 17:14:19	FALSE	Treadmill Running	5.00	470	00:33:22	166	176	3.8	141	150	6:40	6:16	--	--	1.08	8.7	9.3	338	--	329	0.0	326	369	4,688	-9	--	No	00:00:19.0	6	--	00:32:58	00:33:22	--	--
Running	2026-02-19 16:52:23	FALSE	Tuta Running	5.01	459	00:32:58	160	173	3.2	148	150	6:35	5:02	22	25	1.03	8.9	9.2	321	6:36	344	0.0	341	465	4,852	-8	--	No	00:00:04.0	6	--	00:32:54	00:32:58	130	140
Treadmill Running	2026-02-14 15:25:14	FALSE	Treadmill Running	6.50	646	00:45:17	165	179	3.7	138	151	6:58	6:08	--	--	1.06	8.6	9.1	335	--	327	0.0	311	376	6,248	-10	--	No	00:04:18.0	7	--	00:44:49	00:45:17	--	--
Treadmill Running	2026-02-12 16:43:56	FALSE	Treadmill Running	6.06	614	00:46:22	161	177	3.6	130	240	7:39	6:08	--	--	1.00	8.3	8.3	335	--	306	0.0	292	373	6,040	-11	--	No	00:00:55.1	7	--	00:45:31	00:46:22	--	--
Treadmill Running	2026-02-09 18:31:49	FALSE	Treadmill Running	6.00	596	00:46:06	162	176	3.8	131	240	7:41	6:15	--	--	0.99	8.3	8.2	338	--	302	0.0	259	352	6,010	-13	--	No	00:06:22.2	6	--	00:45:13	00:46:06	--	--
Treadmill Running	2026-02-07 17:51:52	FALSE	Treadmill Running	6.00	584	00:46:20	158	177	4.0	130	223	7:43	6:31	--	--	0.98	8.2	8.1	340	--	294	0.0	251	348	6,022	-13	--	No	00:06:45.1	6	--	00:45:44	00:46:20	--	--
Treadmill Running	2026-02-04 16:53:56	FALSE	Treadmill Running	6.00	584	00:46:42	159	177	4.0	131	231	7:47	6:20	--	--	0.98	8.2	8.1	334	--	289	0.0	253	357	6,090	-13	--	No	00:00:19.9	7	--	00:46:13	00:46:42	--	--
Treadmill Running	2026-02-02 16:44:10	FALSE	Treadmill Running	6.00	618	00:46:15	161	183	4.3	132	226	7:43	5:40	--	--	1.01	8.2	8.3	330	--	311	0.0	271	405	6,262	-12	--	No	00:02:14.2	7	--	00:45:35	00:46:21	--	--
Running	2025-11-12 16:48:51	FALSE	Tuta Running	6.23	559	00:40:05	161	175	3.4	148	152	6:26	5:48	32	36	1.05	8.9	9.3	318	6:26	354	0.0	351	448	5,940	-10	--	No	00:01:28.4	7	--	00:40:04	00:40:05	127	139
Running	2025-11-11 16:58:24	FALSE	Tuta Running	10.31	916	01:06:08	161	171	3.8	148	150	6:05	5:50	53	57	1.06	8.8	9.3	320	6:05	392	0.0	350	432	9,756	-17	--	No	00:01:58.5	11	--	01:06:06	01:06:08	127	140
Running	2025-11-09 11:45:20	FALSE	Tuta Running	7.17	648	00:45:18	168	180	3.8	148	153	6:19	5:18	36	39	1.06	8.8	9.3	317	6:20	363	0.0	360	452	6,752	-12	--	No	00:01:05.2	8	--	00:45:16	00:45:49	130	140
Treadmill Running	2025-11-05 16:49:00	FALSE	Treadmill Running	7.40	655	00:45:02	167	180	4.1	147	150	6:05	5:23	--	--	1.12	8.6	9.6	318	--	359	0.0	358	435	6,622	-11	--	No	00:02:23.5	8	--	00:45:01	00:45:02	--	--
Running	2025-11-02 08:11:59	FALSE	Tuta Running	6.03	452	00:32:04	162	173	3.3	149	153	6:22	5:56	22	23	1.05	8.8	9.2	315	6:24	355	0.0	351	435	4,784	-11	--	No	00:00:10.2	6	--	00:32:03	00:32:20	128	138
Running	2025-11-01 09:30:43	FALSE	Tuta Running	6.30	569	00:40:02	164	176	3.5	148	152	6:21	5:15	32	31	1.06	8.8	9.3	317	6:21	355	0.0	353	442	5,920	-12	--	No	00:01:57.0	7	--	00:39:56	00:40:20	129	140
Treadmill Running	2025-10-26 16:16:16	FALSE	Treadmill Running	9.50	841	01:00:05	163	178	4.1	143	200	6:19	5:50	--	--	1.11	8.6	9.6	328	--	347	0.0	344	385	8,574	-15	--	No	00:03:12.3	10	--	00:59:28	01:00:05	--	--
Treadmill Running	2025-10-24 18:10:02	FALSE	Treadmill Running	7.20	649	00:45:33	164	180	4.0	142	150	6:20	5:15	--	--	1.08	8.7	9.4	333	--	333	0.0	331	451	6,468	-11	--	No	00:06:19.2	7	--	00:45:05	00:45:33	--	--
Running	2025-10-21 09:55:44	FALSE	Tuta Running	9.32	842	01:00:06	165	174	3.9	147	150	6:37	5:55	42	44	1.06	8.8	9.3	322	6:26	351	0.0	349	446	8,835	-19	--	No	00:02:06.2	10	--	01:00:03	01:00:06	130	141
Running	2025-10-18 09:34:13	FALSE	Tuta Running	7.33	662	00:45:18	167	178	3.9	148	152	6:11	5:14	36	36	1.09	8.7	9.5	315	6:11	366	0.0	364	442	5,712	-15	--	No	00:01:58.0	8	--	00:45:15	00:45:18	130	140
Running	2025-10-14 09:58:50	FALSE	Tuta Running	6.29	573	00:40:10	165	175	3.6	147	150	6:23	5:32	32	35	1.06	8.8	9.3	320	6:23	353	0.0	351	428	5,930	-12	--	No	00:01:48.4	7	--	00:40:08	00:40:26	129	140
Treadmill Running	2025-10-12 16:44:29	FALSE	Treadmill Running	9.30	786	01:00:04	155	168	3.5	143	240	6:28	6:20	--	--	1.08	8.7	9.4	331	--	336	0.0	334	361	8,558	-16	--	No	00:01:40.6	10	--	00:59:45	01:00:04	--	--
Running	2025-10-07 17:50:19	FALSE	Tuta Running	7.02	630	00:45:12	159	171	3.5	146	150	6:26	4:47	41	44	1.06	8.8	9.3	323	6:26	353	0.0	350	476	6,668	-11	--	No	00:00:09.7	8	--	00:45:08	00:45:20	129	141
Treadmill Running	2025-10-05 14:42:17	FALSE	Treadmill Running	8.00	729	00:51:41	161	177	4.7	143	155	6:28	6:12	--	--	1.11	8.6	9.6	329	--	345	0.0	343	361	7,370	-15	--	No	00:01:08.7	9	--	00:51:21	00:51:41	--	--
Running	2025-10-01 21:35:20	FALSE	Tuta Running	6.04	542	00:39:04	158	171	3.6	146	150	6:28	5:40	44	43	1.06	8.8	9.3	324	6:27	354	0.0	350	460	5,690	-10	--	No	00:00:16.3	7	--	00:39:02	00:39:04	128	140
Treadmill Running	2025-09-26 18:12:55	FALSE	Treadmill Running	7.10	656	00:45:06	155	171	3.5	142	200	6:21	6:13	--	--	1.09	8.7	9.5	332	--	337	0.0	334	351	6,384	-10	--	No	00:06:17.6	7	--	00:44:38	00:45:06	--	--
Running	2025-09-24 17:26:16	FALSE	Qdicik Running	7.77	709	00:50:04	165	177	4.0	147	152	6:27	5:52	17	18	1.05	8.8	9.3	321	6:27	350	0.0	349	475	7,364	-15	--	No	00:04:55.3	8	--	00:50:02	00:50:04	2	7
Treadmill Running	2025-09-22 17:27:15	FALSE	Treadmill Running	7.00	638	00:45:15	160	177	4.1	143	154	6:28	6:15	--	--	1.07	8.8	9.4	328	--	336	0.0	332	375	6,480	-10	--	No	00:05:37.7	7	--	00:44:39	00:45:15	--	--
Treadmill Running	2025-09-20 17:11:43	FALSE	Treadmill Running	6.20	570	00:40:23	161	177	4.4	143	156	6:31	5:11	--	--	1.12	8.6	9.6	326	--	352	0.0	349	444	5,772	-16	--	No	00:03:05.2	7	--	00:40:01	00:40:23	--	--
Running	2025-09-16 07:56:30	FALSE	Qdicik Running	7.01	644	00:45:24	166	177	4.2	147	152	6:29	6:11	10	11	1.04	8.9	9.3	320	6:29	348	0.0	346	488	6,698	-18	--	No	00:00:03.5	8	--	00:45:22	00:45:24	2	10
Running	2025-09-07 18:20:09	FALSE	Kuaidao Running	7.04	647	00:45:08	168	180	3.6	149	155	6:24	6:00	18	17	1.04	8.8	9.2	315	6:24	353	0.0	351	421	6,720	-3	--	No	00:00:17.0	8	--	00:45:06	00:45:08	-1	4
Running	2025-09-06 07:08:41	FALSE	Kuaidao Running	10.07	918	01:04:52	164	178	3.9	149	153	6:27	5:54	4	2	1.04	8.9	9.2	317	6:27	344	0.0	343	400	9,622	-20	--	No	00:00:24.8	11	--	01:04:47	01:04:57	0	3

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5. Data Source and Collection

The dataset consists of self-collected running activity records. Each record corresponds to a single running session tracked using a wearable device.

- Data is non-public and personal
- No sensitive personal information is included
- Environmental classification will be inferred using available indicators (e.g., GPS presence or activity labels)

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6. Methodology

6.1 Data Preparation

- Cleaning missing and inconsistent values
- Formatting timestamps and ordering data chronologically
- Detecting inactivity periods (breaks)
- Labeling running environment (treadmill vs outdoor)

6.2 Feature Engineering

- Weekly training frequency
- Rolling averages for pace
- Performance indicators (e.g., best pace over time)
- Time since last run
- Break duration

6.3 Exploratory Data Analysis

- Time-series visualization of performance trends
- Distribution of pace and heart rate

- Comparison of different training frequency levels
- Visualization of performance before and after breaks

6.4 Hypothesis Testing

- Regression analysis between training frequency and performance
- Comparison of performance before and after inactivity periods
- Correlation analysis between break duration and recovery time
- Statistical comparison of treadmill vs outdoor runs

6.5 Machine Learning Analysis

Machine learning models will be used to analyze and predict performance patterns:

- Regression models: Linear Regression, Random Forest, Gradient Boosting
- Prediction target: Running performance (e.g., pace improvement)
- Evaluation metrics: RMSE, R^2
- Optional classification task: Predict whether performance improves after a given period

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7. Expected Results

- Positive relationship between training frequency and performance improvement
- Significant performance decline after long breaks
- Gradual recovery after returning to training
- Observable differences between treadmill and outdoor running performance

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Repository Structure

01_data_preprocessing.ipynb
02_eda.ipynb
03_hypothesis_testing.ipynb
04_ml_models.ipynb